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END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: ES-102**Subject: Programming in 'C'****Time: 3 Hours****Maximum Marks: 75**

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
Q1	Answer all the following questions briefly:	(5×3=15)
(a)	What is Token? What are the different types of token available in C.	(3)
(b)	Enlist the significance of library functions.	(3)
(c)	What are actual parameters and formal parameters? Illustrate with example.	(3)
(d)	Describe header file and its usage in 'c' programming.	(3)
(e)	Differentiate between put() , putchar() and puts() .	(3)
UNIT - I		
Q2 (a)	Write an algorithm find angle between hour and minutes hand of a clock at a given time.	(7)
(b)	Explain the bit wise AND, OR and NOT operation with suitable example. Mention the role of shift operators.	(8)
Q3 (a)	What is type conversion? Explain two types of conversion with examples.	(7)
(b)	Define pseudocode. Write a pseudocode to find the area of triangle when we know the lengths of all three of its sides.	(8)
UNIT - II		
Q4 (a)	Write a C program that takes three coefficients (a, b and c) of a quadratic equation: $(ax^2 + bx + c)$ as input and compute all possible roots and print them with appropriate messages.	(8)
(b)	What is array? Explain the declaration and initialization of one dimensional and two dimensional arrays with an example.	(7)
Q5 (a)	Shivam is working on a project which requires returning multiple values from a function. But a return statement can only return single value from a function. How the function should be implemented so that multiple values can be returned by Shivam. Explain with suitable example.	(8)
(b)	Write a C program to eliminate multiple spaces from a sentence and make it single.	(7)
UNIT - III		
Q6 (a)	What is a pointer? Write a C program to find the sum and mean of all elements in an array using pointer.	(6)
(b)	Write short note on: i) Dynamic memory allocation ii) Enumeration iii) Self referential structures	(3×3=9)
Q7 (a)	Write a program in C to count the number of words and characters in a file.	(7)
(b)	Define a structure data type named Time containing 3 members called hr, min and sec. Develop a program that would assign values to the individual members and display the time in the form 16:40:30 and also add structure t1 and t2.	(8)
UNIT - IV		
Q8 (a)	What is meant by standard C libraries? What is the purpose of the stdio.h library in C?	(5)
(b)	What do you mean by Sorting? Write a program to sort a list of 'n' numbers in descending order using bubble sort.	(10)
Q9 (a)	Write an algorithm and develop a C program to search an integer from N numbers using binary searching technique.	(5)

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Q. No.	Question	Marks
(b)	What are some key features introduced in the C99 extensions? How do C99 extensions improve the functionality of the C programming language?	(10)

**** End of Question Paper ****